

WC Groupstage Predictions

Model Explanations

ELO (Since Last World Cup): This model uses ELO ratings calculated from international match results since the last World Cup (November 2022). It predicts the probability of a home team winning based on the difference in current ELO ratings between the two teams.

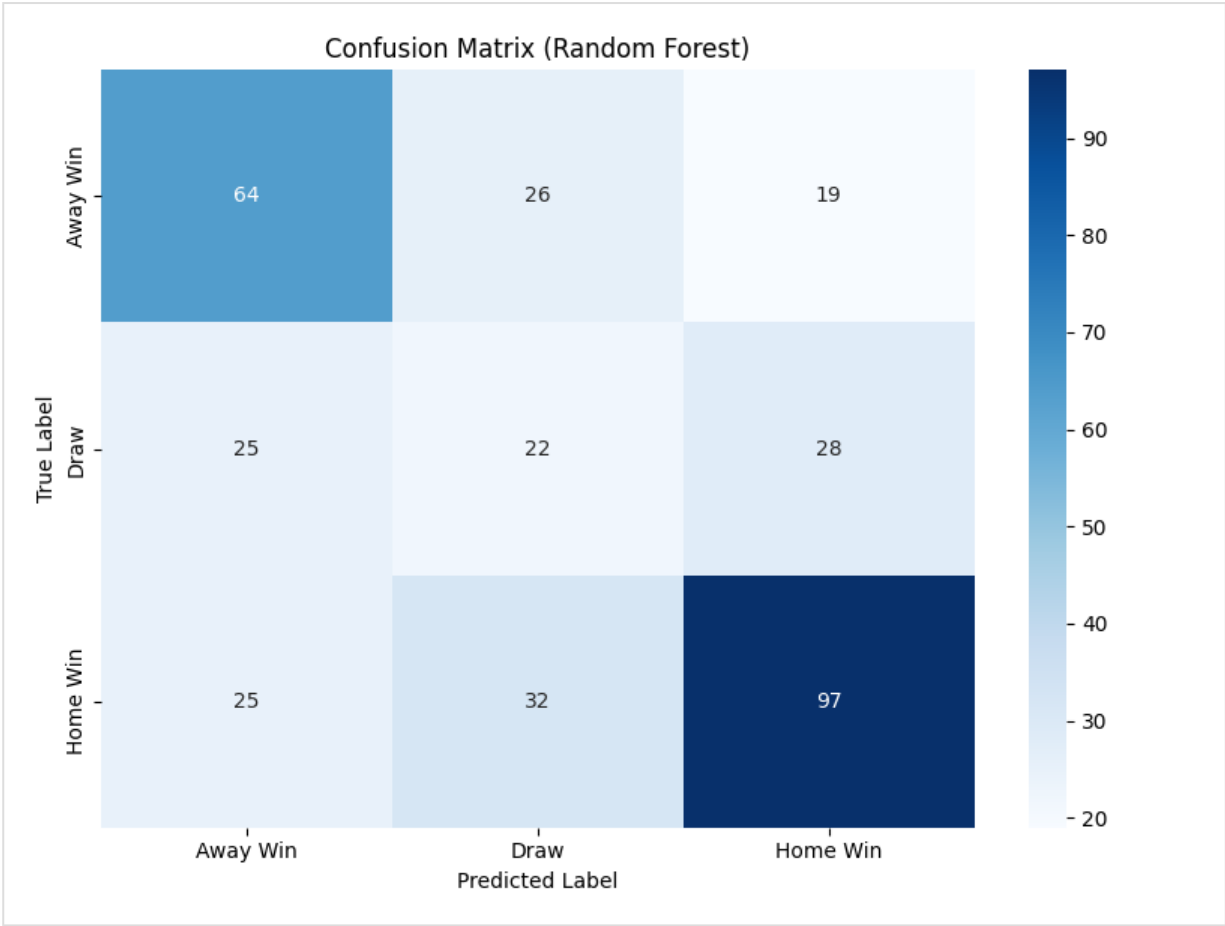
ELO (Historic): Similar to the above, but based on long-term historic ELO ratings, providing a broader view of team strength over time. It predicts win probabilities based on these established historic ratings.

Random Forest Classifier: A powerful ensemble learning method that builds multiple decision trees during training and outputs the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. It is trained on ELO differences, home advantage, and win streaks to predict home win, draw, or away win probabilities.

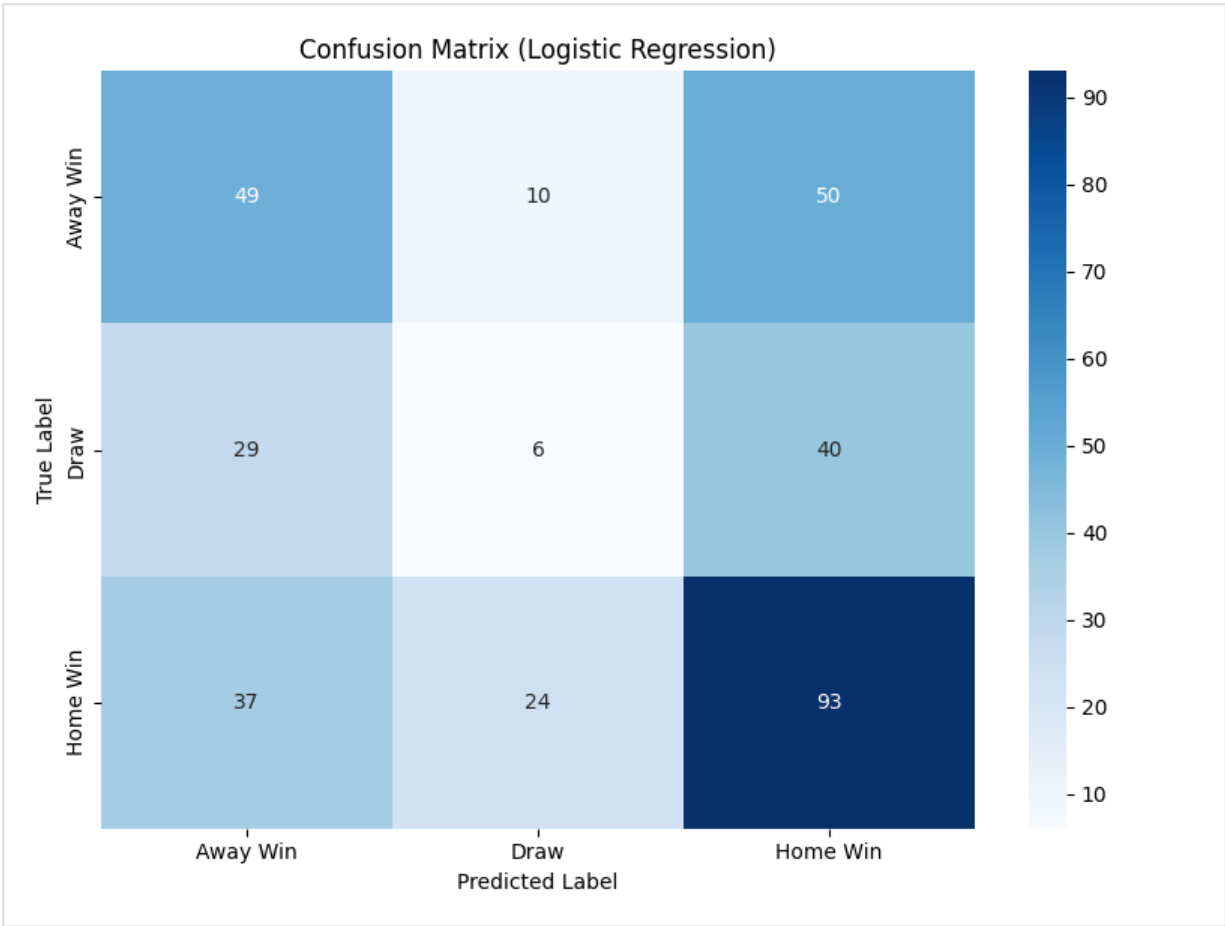
Logistic Regression: A linear model used for predicting the probability of a binary or multi-class outcome. Despite its name, it is a classification algorithm. It uses a logistic function to model the probability of a match outcome (home win, draw, or away win) based on the input features like ELO differences and win streaks.

Model Performance: Confusion Matrices

Random Forest Classifier



Logistic Regression



World Cup Group Stage Predictions

This table presents the enriched group stage match data, including ELO-based probabilities and model predictions from both the Random Forest and Logistic Regression classifiers. The 'home_win_RF', 'tie_RF', and 'home_lose_RF' columns represent the probabilities for a home win, draw, or home loss (away win) as predicted by the Random Forest model, respectively. Similarly, 'home_win_LR', 'tie_LR', and 'home_lose_LR' represent the probabilities from the Logistic Regression model.

kickoff_u tc	home_na me	away_na me	model_h ome_pct	model_dr aw_pct	model_a way_pct	home_elo _last4yrs	away_elo _last4yrs	home_wi n_elo_las t4yrs	home_elo _historic	away_elo _historic	home_wi n_elo_his toric	elo_differ ence	home_los e_RF	tie_RF	home_wi n_RF	home_los e_LR	tie_LR	home_wi n_LR
11062026	Mexico	South Africa	88	7	5	1563.07	1500.00	0.59	1835.00	1500.00	0.87	63.07	0.00	0.82	0.18	0.35	0.40	0.26
12062026	South Korea	Czechia	49	25	26	1532.36	1500.00	0.55	1500.00	1731.00	0.21	32.36	0.03	0.53	0.44	0.38	0.39	0.23
12062026	Canada	Bosnia & Herzegovi na	79	13	8	1479.34	1426.27	0.58	1802.00	1500.00	0.85	53.06	0.61	0.35	0.03	0.35	0.39	0.25
13062026	United States	Paraguay	70	17	13	1494.48	1464.52	0.54	1500.00	1833.00	0.13	29.95	0.23	0.55	0.22	0.38	0.39	0.23
13062026	Qatar	Switzerlan d	8	13	79	1454.57	1504.78	0.43	1427.00	1897.00	0.06	-50.21	0.72	0.26	0.02	0.45	0.38	0.17
13062026	Brazil	Morocco	44	26	30	1560.29	1549.90	0.51	1979.00	1830.00	0.70	10.39	0.65	0.20	0.15	0.40	0.39	0.21
14062026	Haiti	Scotland	7	12	81	1472.07	1445.34	0.54	1542.00	1790.00	0.19	26.73	0.84	0.07	0.10	0.38	0.39	0.23
14062026	Australia	Türkiye	27	25	48	1492.95	1476.38	0.52	1774.00	1880.00	0.35	16.57	0.58	0.31	0.11	0.39	0.39	0.22
14062026	Germany	Curaçao	95	3	2	1543.67	1500.00	0.56	1910.00	1500.00	0.91	43.67	0.26	0.00	0.74	0.37	0.39	0.24
14062026	Netherlan ds	Japan	57	23	20	1511.65	1545.76	0.45	1959.00	1878.00	0.61	-34.11	0.68	0.25	0.06	0.44	0.38	0.18
14062026	Ivory Coast	Ecuador	23	24	53	1529.94	1540.98	0.48	1500.00	1933.00	0.08	-11.04	0.36	0.53	0.11	0.42	0.39	0.19
15062026	Sweden	Tunisia	49	25	26	1456.46	1504.67	0.43	1660.00	1641.00	0.53	-48.21	0.73	0.23	0.04	0.45	0.38	0.17

kickoff_u tc	home_na me	away_na me	model_h ome_pct	model_dr aw_pct	model_a way_pct	home_elo _last4yrs	away_elo _last4yrs	home_wi n_elo_las t4yrs	home_elo _historic	away_elo _historic	home_wi n_elo_his toric	elo_differ ence	home_los e_RF	tie_RF	home_wi n_RF	home_los e_LR	tie_LR	home_wi n_LR
15062026	Spain	Cape Verde	94	4	2	1575.52	1479.02	0.64	2171.00	1500.00	0.98	96.50	0.65	0.19	0.16	0.32	0.40	0.29
15062026	Belgium	Egypt	63	21	16	1533.18	1487.29	0.57	1849.00	1591.00	0.82	45.89	0.58	0.11	0.31	0.36	0.39	0.24
15062026	Saudi Arabia	Uruguay	6	10	84	1430.98	1514.91	0.38	1500.00	1890.00	0.10	-83.93	0.67	0.31	0.03	0.48	0.37	0.15
16062026	Iran	New Zealand	92	5	3	1479.50	1407.69	0.60	1754.00	1500.00	0.81	71.81	0.50	0.42	0.07	0.34	0.39	0.27
16062026	France	Senegal	56	23	21	1565.04	1492.85	0.60	2062.00	1803.00	0.82	72.19	0.64	0.21	0.16	0.34	0.40	0.26
16062026	Iraq	Norway	12	17	71	1456.17	1483.72	0.46	1582.00	1922.00	0.12	-27.55	0.65	0.29	0.06	0.43	0.38	0.19
17062026	Argentina	Algeria	70	17	13	1579.45	1550.16	0.54	2113.00	1726.00	0.90	29.30	0.41	0.38	0.22	0.38	0.40	0.22
17062026	Austria	Jordan	81	12	7	1561.24	1465.54	0.63	1818.00	1687.00	0.68	95.70	0.76	0.17	0.06	0.32	0.40	0.29
17062026	Portugal	DR Congo	82	11	7	1578.42	1463.06	0.66	1976.00	1500.00	0.94	115.36	0.80	0.08	0.12	0.30	0.40	0.30
17062026	England	Croatia	47	26	27	1504.62	1500.25	0.51	2042.00	1933.00	0.65	4.37	0.26	0.72	0.02	0.40	0.39	0.21
17062026	Ghana	Panama	8	14	78	1403.04	1423.33	0.47	1509.00	1742.00	0.21	-20.29	0.87	0.10	0.03	0.42	0.38	0.20
18062026	Uzbekistan	Colombia	8	13	79	1460.69	1571.18	0.35	1735.00	1998.00	0.18	-110.50	0.94	0.04	0.02	0.50	0.37	0.13
18062026	Czechia	South Africa	62	21	17	1500.00	1500.00	0.50	1731.00	1500.00	0.79	0.00	0.37	0.60	0.03	0.40	0.39	0.21
18062026	Switzerland	Bosnia & Herzegovina	83	11	6	1504.78	1426.27	0.61	1897.00	1500.00	0.91	78.51	0.85	0.03	0.12	0.33	0.39	0.28
18062026	Canada	Qatar	75	15	10	1479.34	1454.57	0.54	1802.00	1427.00	0.90	24.77	0.47	0.31	0.21	0.38	0.39	0.23
19062026	Mexico	South Korea	60	22	18	1563.07	1532.36	0.54	1835.00	1500.00	0.87	30.71	0.15	0.71	0.15	0.38	0.39	0.23
19062026	United States	Australia	62	21	17	1494.48	1492.95	0.50	1500.00	1774.00	0.17	1.53	0.57	0.37	0.06	0.40	0.39	0.21
19062026	Scotland	Morocco	13	17	70	1445.34	1549.90	0.35	1790.00	1830.00	0.44	-104.56	0.86	0.05	0.09	0.49	0.37	0.13
20062026	Brazil	Haiti	96	3	1	1560.29	1472.07	0.62	1979.00	1542.00	0.93	88.21	0.76	0.20	0.04	0.32	0.40	0.28
20062026	Türkiye	Paraguay	57	23	20	1476.38	1464.52	0.52	1880.00	1833.00	0.57	11.86	0.15	0.66	0.20	0.39	0.39	0.22
20062026	Netherlands	Sweden	72	16	12	1511.65	1456.46	0.58	1959.00	1660.00	0.85	55.19	0.79	0.02	0.20	0.35	0.39	0.25

kickoff_u tc	home_na me	away_na me	model_h ome_pct	model_dr aw_pct	model_a way_pct	home_elo _last4yrs	away_elo _last4yrs	home_wi n_elo_las t4yrs	home_elo _historic	away_elo _historic	home_wi n_elo_his toric	elo_differ ence	home_los e_RF	tie_RF	home_wi n_RF	home_los e_LR	tie_LR	home_wi n_LR
20062026	Germany	Ivory Coast	68	18	14	1543.67	1529.94	0.52	1910.00	1500.00	0.91	13.73	0.39	0.37	0.23	0.39	0.39	0.21
21062026	Ecuador	Curaçao	91	6	3	1540.98	1500.00	0.56	1933.00	1500.00	0.92	40.98	0.32	0.00	0.68	0.37	0.39	0.24
21062026	Tunisia	Japan	15	20	65	1504.67	1545.76	0.44	1641.00	1878.00	0.20	-41.08	0.76	0.15	0.09	0.44	0.38	0.17
21062026	Spain	Saudi Arabia	92	5	3	1575.52	1430.98	0.70	2171.00	1500.00	0.98	144.54	0.83	0.01	0.17	0.27	0.40	0.33
21062026	Belgium	Iran	57	23	20	1533.18	1479.50	0.58	1849.00	1754.00	0.63	53.68	0.62	0.06	0.32	0.36	0.39	0.25
21062026	Uruguay	Cape Verde	88	8	4	1514.91	1479.02	0.55	1890.00	1500.00	0.90	35.89	0.63	0.06	0.31	0.37	0.39	0.24
22062026	New Zealand	Egypt	4	6	90	1407.69	1487.29	0.39	1500.00	1591.00	0.37	-79.60	0.67	0.21	0.12	0.47	0.37	0.15
22062026	Argentina	Austria	60	22	18	1579.45	1561.24	0.53	2113.00	1818.00	0.85	18.21	0.53	0.28	0.18	0.39	0.39	0.21
22062026	France	Iraq	91	6	3	1565.04	1456.17	0.65	2062.00	1582.00	0.94	108.87	0.90	0.06	0.05	0.30	0.40	0.30
23062026	Norway	Senegal	24	25	51	1483.72	1492.85	0.49	1922.00	1803.00	0.66	-9.12	0.56	0.40	0.04	0.41	0.39	0.20
23062026	Jordan	Algeria	11	15	74	1465.54	1550.16	0.38	1687.00	1726.00	0.44	-84.62	0.78	0.09	0.13	0.48	0.38	0.15
23062026	Portugal	Uzbekista n	85	10	5	1578.42	1460.69	0.66	1976.00	1735.00	0.80	117.74	0.80	0.09	0.11	0.30	0.40	0.31
23062026	England	Ghana	95	3	2	1504.62	1403.04	0.64	2042.00	1509.00	0.96	101.57	0.85	0.04	0.11	0.31	0.39	0.30
23062026	Panama	Croatia	15	19	66	1423.33	1500.25	0.39	1742.00	1933.00	0.25	-76.92	0.66	0.31	0.04	0.47	0.38	0.15
24062026	Colombia	DR Congo	75	15	10	1571.18	1463.06	0.65	1998.00	1500.00	0.95	108.12	0.83	0.14	0.04	0.30	0.40	0.30
24062026	Switzerlan d	Canada	43	26	31	1504.78	1479.34	0.54	1897.00	1802.00	0.63	25.44	0.53	0.20	0.28	0.38	0.39	0.23
24062026	Bosnia & Herzegovi na	Qatar	31	27	42	1426.27	1454.57	0.46	1500.00	1427.00	0.60	-28.29	0.86	0.05	0.08	0.43	0.38	0.19
24062026	Morocco	Haiti	95	3	2	1549.90	1472.07	0.61	1830.00	1542.00	0.84	77.82	0.74	0.18	0.08	0.33	0.40	0.27
24062026	Scotland	Brazil	10	14	76	1445.34	1560.29	0.34	1790.00	1979.00	0.25	-114.95	0.92	0.03	0.06	0.50	0.37	0.13
25062026	South Africa	South Korea	11	16	73	1500.00	1532.36	0.45	1500.00	1500.00	0.50	-32.36	0.44	0.21	0.35	0.44	0.38	0.18
25062026	Czechia	Mexico	12	17	71	1500.00	1563.07	0.41	1731.00	1835.00	0.35	-63.07	0.72	0.01	0.27	0.46	0.38	0.16
25062026	Curaçao		6	10	84	1500.00	1529.94	0.46	1500.00	1500.00	0.50	-29.94	0.99	0.01	0.01	0.43	0.39	0.18

kickoff_u tc	home_na me	away_na me	model_h ome_pct	model_dr aw_pct	model_a way_pct	home_elo _last4yrs	away_elo _last4yrs	home_wi n_elo_las t4yrs	home_elo _historic	away_elo _historic	home_wi n_elo_his toric	elo_differ ence	home_los e_RF	tie_RF	home_wi n_RF	home_los e_LR	tie_LR	home_wi n_LR
		Ivory Coast																
25062026	Ecuador	Germany	23	25	52	1540.98	1543.67	0.50	1933.00	1910.00	0.53	-2.69	0.62	0.28	0.10	0.41	0.39	0.20
25062026	Tunisia	Netherlan ds	7	12	81	1504.67	1511.65	0.49	1641.00	1959.00	0.14	-6.98	0.60	0.37	0.03	0.41	0.39	0.20
25062026	Japan	Sweden	53	24	23	1545.76	1456.46	0.63	1878.00	1660.00	0.78	89.30	0.72	0.14	0.13	0.32	0.40	0.28
26062026	Türkiye	United States	25	25	50	1476.38	1494.48	0.47	1880.00	1500.00	0.90	-18.09	0.73	0.22	0.05	0.42	0.39	0.19
26062026	Paraguay	Australia	28	26	46	1464.52	1492.95	0.46	1833.00	1774.00	0.58	-28.42	0.88	0.11	0.01	0.43	0.38	0.19
26062026	Norway	France	13	18	69	1483.72	1565.04	0.39	1922.00	2062.00	0.31	-81.32	0.80	0.13	0.07	0.48	0.38	0.15
26062026	Senegal	Iraq	82	11	7	1492.85	1456.17	0.55	1803.00	1582.00	0.78	36.68	0.62	0.28	0.10	0.37	0.39	0.24
27062026	Cape Verde	Saudi Arabia	29	27	44	1479.02	1430.98	0.57	1500.00	1500.00	0.50	48.04	0.74	0.21	0.06	0.36	0.39	0.25
27062026	Uruguay	Spain	22	24	54	1514.91	1575.52	0.41	1890.00	2171.00	0.17	-60.61	0.66	0.17	0.17	0.46	0.38	0.16
27062026	New Zealand	Belgium	1	3	96	1407.69	1533.18	0.33	1500.00	1849.00	0.12	-125.49	0.80	0.16	0.04	0.51	0.37	0.12
27062026	Egypt	Iran	31	26	43	1487.29	1479.50	0.51	1591.00	1754.00	0.28	7.79	0.20	0.65	0.15	0.40	0.39	0.21
27062026	Panama	England	10	15	75	1423.33	1504.62	0.39	1742.00	2042.00	0.15	-81.28	0.55	0.44	0.01	0.47	0.37	0.15
27062026	Croatia	Ghana	92	5	3	1500.25	1403.04	0.64	1933.00	1509.00	0.92	97.21	0.74	0.15	0.11	0.31	0.39	0.29
27062026	Colombia	Portugal	28	25	47	1571.18	1578.42	0.49	1998.00	1976.00	0.53	-7.24	0.75	0.11	0.14	0.42	0.39	0.19
27062026	DR Congo	Uzbekista n	42	26	32	1463.06	1460.69	0.50	1500.00	1735.00	0.21	2.37	0.05	0.90	0.05	0.40	0.39	0.21
28062026	Algeria	Austria	28	25	47	1550.16	1561.24	0.48	1726.00	1818.00	0.37	-11.08	0.80	0.06	0.15	0.42	0.39	0.19
28062026	Jordan	Argentina	3	5	92	1465.54	1579.45	0.34	1687.00	2113.00	0.08	-113.91	0.88	0.05	0.08	0.50	0.37	0.13